TELECOM NETWORK OPERATIONS

Knowledge Base (SOP)

AI-Powered Incident Response & Troubleshooting Guide

Version	2.0
Document Type	Internal Operational Knowledge Base (RAG Source)
Primary Audience	NOC Engineers and the AI Telecom Incident Copilot (RAG retrieval layer)
Supported Root Causes	RAN Congestion, Backhaul/Transport Issue, Radio Coverage Degradation, Interference/Poor Signal Quality, Mobility/Handover Issue, Suspected Cell Outage/Severe Service Drop

Purpose. This knowledge base supports Retrieval-Augmented Generation (RAG) for AI-assisted telecom incident analysis. It is written for two readers at once: NOC engineers performing troubleshooting, and the AI Telecom Incident Copilot retrieving context by semantic similarity. The document complements — and does not duplicate — the existing rule-based recommendation agent. Each root-cause entry provides operational context, a troubleshooting workflow, validation checks, network-slice-specific considerations, escalation guidance, and expected operational impact. The RCA engine's root-cause taxonomy and confidence scoring framework are authoritative and are not restated here.

Contents

1. RAN Congestion

Radio resource saturation limiting scheduling efficiency in the serving cell

Overview

RAN Congestion occurs when radio resources in a serving cell become saturated and the cell cannot efficiently schedule user traffic. RAN Congestion is most common during peak traffic periods, large-scale events, or when subscriber demand exceeds available radio capacity. RAN Congestion typically affects PRB utilization, throughput, latency, and active user count at the same time.

RCA Detection Logic

The RCA engine flags RAN Congestion when the following KPIs move together: high PRB utilization, high active user count, low throughput, increased latency, and increased packet loss. Confidence in a RAN Congestion classification increases when multiple congestion indicators are observed concurrently rather than in isolation. Confidence scoring itself is derived from the RCA scoring framework and is not repeated in this entry.

Typical Symptoms

- High PRB utilization sustained during busy-hour traffic
- Reduced user throughput despite otherwise stable radio conditions
- Increased scheduling delay for user traffic
- Packet loss concentrated during busy-hour periods
- High number of simultaneously active users on the serving cell

Primary KPIs

PRB Utilization	**Throughput**	**Packet Loss**	**Latency**	**Active Users**

Network Slice Considerations

eMBB — RAN Congestion on eMBB slices primarily reduces throughput. Confirm PRB allocation fairness and load balancing across neighboring cells before assuming a hard capacity limitation.

URLLC — RAN Congestion on URLLC slices threatens latency guarantees more than throughput. Verify scheduler prioritization for latency-sensitive traffic first.

mMTC — RAN Congestion on mMTC slices typically shows up as random access congestion and control channel saturation rather than PRB congestion. Check access success rates for the affected cell.

Immediate Actions

- Verify PRB utilization trend for the affected cell against neighboring cells
- Review neighboring cell utilization to confirm whether congestion is isolated or cluster-wide
- Check scheduler performance for signs of resource starvation
- Verify traffic balancing across sectors and carriers
- Review transport utilization to rule out a backhaul contribution to the congestion pattern

Short-Term Actions

- Enable or adjust load balancing between the affected cell and its neighbors
- Optimize traffic steering to redistribute active users
- Review scheduler configuration for the affected cell
- Optimize resource allocation for known high-demand time windows

Long-Term Actions

- Evaluate capacity expansion for the affected cell or cluster
- Add carrier or spectrum resources where feasible
- Reassess sectorization for persistently congested sites

Escalation

Primary Team	Radio Optimization
Secondary Team	Capacity Planning
Condition	Escalate to Transport Operations if backhaul utilization is found to be a contributing factor.

Expected Outcome

> Reduced PRB utilization, improved throughput, lower latency, and a better-balanced load across the affected cell and its neighbors.

2. Backhaul / Transport Issue

Impaired or congested transport links between the radio network and core network

Overview

Backhaul/Transport Issues occur when transport links connecting the radio network to the core network become impaired or congested. A Backhaul/Transport Issue is distinguished from RAN Congestion because the radio interface itself is not the bottleneck; the impairment sits in the transport path.

RCA Detection Logic

The RCA engine prioritizes a Backhaul/Transport classification when packet loss is high or very high, latency is high or very high, throughput is low, and PRB utilization remains below congestion thresholds. Low PRB utilization combined with poor throughput and latency is the key differentiator between a transport problem and RAN Congestion.

Typical Symptoms

- High latency not explained by radio conditions
- Packet loss independent of PRB utilization
- Throughput degradation despite low radio-side congestion
- Stable radio utilization despite poor end-user service quality

Primary KPIs

Latency	**Packet Loss**	**Throughput**	**PRB Utilization**

Network Slice Considerations

eMBB	Backhaul/Transport Issues on eMBB slices reduce throughput. Validate transport link capacity and interface utilization before taking any radio-side action.
URLLC	Backhaul/Transport Issues on URLLC slices directly threaten latency and jitter guarantees. Confirm redundant path availability for latency-sensitive traffic.
mMTC	Backhaul/Transport Issues on mMTC slices can degrade signaling reliability for massive device connectivity even when data volumes are low.

Immediate Actions

- Check transport link health for the affected cell site
- Verify interface errors on transport equipment
- Review packet drops at transport interfaces
- Validate routing paths between the cell site and the core network

Short-Term Actions

- Reroute traffic over an alternate transport path where available
- Balance transport utilization across available links
- Monitor latency trends following any transport-side change

Long-Term Actions

- Increase transport capacity for chronically congested links
- Upgrade backhaul infrastructure at affected sites
- Improve transport redundancy to reduce single points of failure

Escalation

Primary Team	Transport Operations
Secondary Team	Core Network Operations

Expected Outcome

> Reduced latency, lower packet loss, and restored service stability without requiring radio-side intervention.

3. Radio Coverage Degradation

Insufficient signal strength to maintain reliable service

Overview

Radio Coverage Degradation occurs when radio signal strength is insufficient to maintain reliable service across part of a cell's footprint. Coverage Degradation is fundamentally a signal-strength problem rather than a capacity or interference problem.

RCA Detection Logic

The RCA engine identifies Coverage Degradation through weak RSRP, poor throughput, high packet loss, and poor RSRQ. Weak RSRP is the primary discriminator that separates Coverage Degradation from Interference, where RSRP typically remains acceptable.

Typical Symptoms

- Weak RSRP across affected areas of the cell footprint
- Poor throughput correlated with weak signal strength
- High packet loss concentrated at the cell edge
- Poor RSRQ accompanying weak RSRP

Primary KPIs

RSRP	**RSRQ**	**Throughput**	**Packet Loss**

Network Slice Considerations

eMBB	Coverage Degradation on eMBB slices should be prioritized wherever throughput coverage across the cell footprint is impacted.
URLLC	Coverage Degradation on URLLC slices threatens the reliability of low-latency radio links and should be treated with urgency in areas serving latency-sensitive traffic.
mMTC	Coverage Degradation on mMTC slices reduces broad device accessibility and can lower connection success rates for IoT devices at cell edge.

Immediate Actions

- Review RSRP distribution across the affected cell footprint
- Verify antenna configuration against the intended design
- Check antenna tilt and azimuth for unintended drift
- Inspect for physical obstructions near the site

Short-Term Actions

- Adjust antenna tilt or azimuth where misconfiguration is confirmed

- Validate coverage improvement after any antenna adjustment
- Review neighboring cell coverage overlap to confirm no residual coverage gap remains

Long-Term Actions

- Plan coverage optimization for the affected area
- Evaluate antenna reconfiguration or upgrade
- Assess additional site deployment where a coverage gap cannot be closed by existing sites

Escalation

Primary Team	RF Optimization
Secondary Team	Field Maintenance

Expected Outcome

> Improved RSRP and RSRQ across the affected footprint, higher throughput at the cell edge, and reduced packet loss.

4. Interference / Poor Signal Quality

Degraded signal quality despite acceptable signal strength

Overview

Interference occurs when radio signal quality degrades despite signal strength remaining acceptable. Interference is distinguished from Coverage Degradation because RSRP is not severely impacted while RSRQ deteriorates.

RCA Detection Logic

The RCA engine emphasizes poor or very poor RSRQ, high packet loss, and high latency, combined with RSRP that is not severely degraded. This RSRP/RSRQ relationship is the key differentiator between Interference and Coverage Degradation.

Typical Symptoms

- Poor RSRQ despite acceptable RSRP
- High packet loss not explained by weak signal strength
- Increased latency correlated with signal quality degradation

Primary KPIs

RSRQ	**Packet Loss**	**Latency**	**RSRP**

Network Slice Considerations

eMBB	Interference on eMBB slices reduces effective throughput even when reported signal strength appears normal. Review RSRQ alongside throughput trends.
URLLC	Interference on URLLC slices increases packet delay variation and can violate latency guarantees even without a drop in RSRP.
mMTC	Interference on mMTC slices can raise random access failures and degrade signaling reliability for massive device connectivity.

Immediate Actions

- Review neighboring cell interference sources
- Verify PCI (Physical Cell Identity) planning for conflicts
- Review frequency allocation for overlap or reuse issues
- Investigate external interference sources near the affected site

Short-Term Actions

- Adjust PCI or frequency assignment where a conflict is confirmed
- Coordinate with neighboring cells to reduce mutual interference

- Monitor RSRQ trend after any interference mitigation step

Long-Term Actions

- Plan RF optimization for the affected cluster
- Conduct frequency replanning where recurring interference is confirmed
- Implement interference mitigation measures at the site or cluster level

Escalation

Primary Team	RF Optimization
Secondary Team	Spectrum Engineering

Expected Outcome

> Improved RSRQ, reduced packet loss, lower latency, and more consistent signal quality across the affected cell.

5. Mobility / Handover Issue

Degraded service quality during transitions between cells

Overview

Mobility/Handover Issues occur when users experience degraded service quality during transitions between cells. Mobility Issues are most visible along cell boundaries and in areas with dense cell overlap.

RCA Detection Logic

The RCA engine identifies Mobility Issues through a high handover count combined with packet loss, increased latency, and reduced throughput. High handover activity is the principal trigger that distinguishes Mobility Issues from other root causes.

Typical Symptoms

- High handover count for the affected cell or cell pair
- Packet loss concentrated around handover events
- Increased latency during cell transitions
- Reduced throughput correlated with handover activity

Primary KPIs

Handover Count	**Packet Loss**	**Latency**	**Throughput**

Network Slice Considerations

eMBB Mobility Issues on eMBB slices reduce throughput during cell transitions, most noticeably for users moving through overlapping coverage areas.

URLLC Mobility Issues on URLLC slices are especially disruptive because handover interruption directly affects latency guarantees.

mMTC Mobility Issues on mMTC slices are typically less latency-sensitive but can still affect connection continuity for mobile IoT devices.

Immediate Actions

- Review handover counters for the affected cell or cell pair
- Verify neighbor relations are correctly configured
- Review mobility parameters against expected values
- Analyze handover failures for a pattern such as ping-pong, too-early, or too-late handover

Short-Term Actions

- Adjust mobility parameters for the affected cell pair
- Correct neighbor relation lists where gaps or errors are found

- Monitor handover success rate after any parameter change

Long-Term Actions

- Plan neighbor list optimization for the affected cluster
- Tune mobility parameters as part of broader RF optimization
- Apply SON (Self-Organizing Network) optimization where available

Escalation

Primary Team	RAN Optimization
Secondary Team	SON Team

Expected Outcome

> Reduced handover count and failures, lower packet loss during transitions, and improved throughput continuity for mobile users.

6. Suspected Cell Outage / Severe Service Drop

Severe service degradation or complete service loss

Overview

Suspected Cell Outage represents severe service degradation or complete service loss requiring immediate operational attention. This category takes priority over other root causes when its detection conditions are met, since it indicates the cell may be unavailable.

RCA Detection Logic

The RCA engine identifies a Suspected Cell Outage through a combination of very low throughput, low active users, low PRB utilization, high latency, and high packet loss. This combination indicates the cell is likely unavailable or severely impaired rather than simply congested or degraded.

Typical Symptoms

- Very low or near-zero throughput
- Low active user count despite expected traffic demand
- Low PRB utilization inconsistent with normal cell activity
- High latency and high packet loss for any traffic that is still served

Primary KPIs

Throughput	**Active Users**	**PRB Utilization**	**Packet Loss**	**Latency**

Network Slice Considerations

eMBB	A Suspected Cell Outage on eMBB slices results in a complete or near-complete loss of throughput for affected users.
URLLC	A Suspected Cell Outage on URLLC slices is critical, since latency-sensitive services lose service continuity entirely.
mMTC	A Suspected Cell Outage on mMTC slices disrupts massive device connectivity and can cause a backlog of failed IoT connection attempts.

Immediate Actions

- Verify cell availability in the network management system
- Review site alarms for power, hardware, or transmission faults
- Check power status at the site
- Verify transmission/transport connectivity to the site
- Review recent configuration changes that may have triggered the outage

Short-Term Actions

- Restore site connectivity where a transport fault is confirmed
- Recover or replace failed hardware where applicable
- Validate service restoration through KPI recovery for the affected cell

Long-Term Actions

- Conduct root cause investigation for the underlying failure
- Replace hardware where a recurring fault is identified
- Schedule preventive maintenance to reduce recurrence

Escalation

Primary Team	Field Operations
Secondary Team	Hardware Maintenance
Condition	Emergency escalation to the Network Operations Center (NOC) for immediate response.

Expected Outcome

> Rapid restoration of service availability and recovery of normal throughput, latency, and packet loss levels for the affected cell.

General Escalation Matrix

Consolidated view of primary and secondary escalation teams by root cause, for quick reference during incident triage.

Root Cause	Primary Team	Secondary Team
RAN Congestion	Radio Optimization	Capacity Planning
Backhaul / Transport Issue	Transport Operations	Core Network Operations
Radio Coverage Degradation	RF Optimization	Field Maintenance
Interference / Poor Signal Quality	RF Optimization	Spectrum Engineering
Mobility / Handover Issue	RAN Optimization	SON Team
Suspected Cell Outage / Severe Service Drop	Field Operations	Hardware Maintenance

Document Conventions

- This SOP complements the rule-based recommendation agent. It does not restate the agent's automated recommendations; it provides the operational context, workflow, and reasoning behind them.
- The RCA engine supports exactly six root causes: RAN Congestion, Backhaul/Transport Issue, Radio Coverage Degradation, Interference/Poor Signal Quality, Mobility/Handover Issue, and Suspected Cell Outage/Severe Service Drop. No other root causes are in scope for this knowledge base.
- Slice-specific guidance is organized by network_slice only (eMBB, URLLC, mMTC). Cell-type-specific guidance is intentionally out of scope for this version.
- This document does not define numeric KPI thresholds, vendor-specific commands, or alarm codes. Threshold and alarm logic reside in the RCA scoring framework and vendor operations manuals respectively.